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Downhole setting assembly with switch module

Abstract

A system for plugging a wellbore includes a plug including an annular sealing element, a setting tool configured to couple with the plug and including an energetic element, the setting tool configured to shift the plug from an initial configuration to a set configuration in response to ignition of the energetic element, and a setting tool initiator including an addressable igniter switch and an igniter, wherein the igniter is configured to ignite the energetic element of the setting tool in response to the igniter switch receiving a firing signal addressed to the igniter switch, and wherein the setting tool initiator is configured to shift the igniter switch from an operational state to an inoperable state in response to the ignition of the igniter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. non-provisional patent application Ser. No. 17/742,185 filed May 11, 2022, entitled “Downhole Setting Assembly with Switch Module”, which claims benefit of U.S. provisional patent application No. 63/187,145 filed May 11, 2021, entitled “Downhole Setting Assembly with

Switch Module,” both of which are hereby incorporated herein by reference in their entirety for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not applicable.

BACKGROUND

(2) During completion operations for a subterranean wellbore, it is conventional practice to perforate the wellbore with perforating guns along with any casing tubulars disposed therein along a targeted hydrocarbon bearing formation to provide a path for formation fluids (e.g., hydrocarbons) to flow into the wellbore. To enhance the productivity of each of typically a great many perforations, the wellbore is divided into a plurality of production zones along the targeted formation where the perforations associated with each zone are enlarged and expanded by hydraulic fracturing sometimes referred to as “fracking”. Each production zone is isolated from the other downhole zones using a sealing device (e.g., a plug, a packer) installed within the wellbore prior to the given production zone being perforated. Generally, both a setting tool and at least one perforating gun assembled along the same tool string are inserted into the wellbore in order to set the sealing device and then perforate the casing in a single trip downhole.

(3) Typically, the sealing device is installed from a downhole end of the tool string using a setting tool and initiator which is sometimes referred to as a “firing head.” The setting tool typically includes an explosive or combustible power charge for shifting the sealing device within the wellbore from an initial configuration in which fluid flow is permitted around the sealing device and a set configuration in which the sealing device plugs the wellbore. With the sealing device in the set configuration the setting tool separates from the set sealing device to permit the setting tool to be pulled back to the surface with the rest of the tool string. The firing head or setting tool initiator typically includes a firing head or igniter switch that is connected to a controller at the surface and to the combustible element through an igniter of the firing head. However, it may be understood that a given sealing device is set before any perforations are created uphole from the intended location of the setting of the given sealing device as the ability to effectively frack any of those uphole perforations is undermined by the open and fracked perforations downhole from the given sealing device. Thus, a failure to properly set the sealing device may result in fluid communication between the perforations formed uphole from the sealing device and the fracked perforations located downhole from the sealing device, preventing the operator from successfully fracking the uphole perforations due to diversion of the fracking fluid into the downhole perforations.

(4) Recognizing that the operator at the surface has a high need to know that the sealing device is fully set and sealing off the downhole zones of the wellbore, the operator may confirm that the sealing device has set by slowly reeling in wireline at the wireline truck and observing an increase in tension on the wireline cable at the surface as the sealing device is gripping to the inside of the casing and then a sudden drop in tension on the wireline cable when the setting tool separates from the now set sealing device. If that characteristic tension change in the wireline cable is not observed, then the operator may pump additional fluid downhole and see if more wireline is drawn out which would suggest that the sealing device has not yet set. Conversely, if the sealing device is set, any further liquid pumping would not push the sealing device farther downhole. While these verification techniques provide some degree of confidence, they are time consuming in an operation in which every additional minute results in added costs. Thus, the industry would value a better, faster, cheaper means for confirming that the plug has set before creating more perforations in a wellbore.

SUMMARY

(5) An embodiment system for plugging a wellbore extending from the surface and into a subterranean earthen formation comprises a plug comprising an annular sealing element and having

an initial configuration configured to permit fluid flow around the plug in the wellbore and a set configuration configured to plug the wellbore whereby fluid flow around the plug in the wellbore is restricted, a setting tool configured to couple with the plug and comprising an energetic element, the setting tool configured to shift the plug from the initial configuration to the set configuration in response to ignition of the energetic element, and a setting tool initiator comprising an addressable igniter switch and an igniter, wherein the igniter is configured to ignite the energetic element of the setting tool in response to the igniter switch receiving a firing signal addressed to the igniter switch, wherein the igniter switch has an operational state in which the igniter switch is configured to receive electrical signals from the surface and an inoperable state in which the igniter switch is not configured receive electrical signals from the surface, and wherein the setting tool initiator is configured to shift the igniter switch from the operational state to the inoperable state in response to the ignition of the igniter. In some embodiments, the igniter switch comprises a digitally addressable switch. In some embodiments, the igniter switch comprises a processor and a memory device including instructions stored therein defining the operation of the igniter switch. In certain embodiments, the setting tool initiator is configured to physically disable the igniter switch in response to the ignition of the igniter. In certain embodiments, the setting tool initiator is configured to expose the igniter switch to combustion products generated by the igniter in response to the ignition of the igniter. In some embodiments, the setting tool initiator is configured to permanently shift the igniter switch from the operational state to the inoperable state. In some embodiments, the setting tool initiator comprises an initiator housing having a switch compartment in which the igniter switch is positioned, and an igniter compartment in which the igniter is located, and wherein a fluid flowpath extends from the igniter compartment to the switch compartment. In certain embodiments, the plugging system comprises a surface controller in signal communication with the igniter switch when the igniter switch is in the operational state and not in signal communication with the igniter switch when the igniter switch is in the inoperable state. In certain embodiments, the setting tool initiator comprises a switch chassis having an interior in which the igniter switch is positioned, and an igniter adapter coupled to the switch chassis and having an interior in which the igniter is positioned.

(6) An embodiment of a setting tool initiator for actuating a setting tool of a plugging system comprises an addressable igniter switch having an operational state in which the igniter switch is configured to receive and transmit electrical signals and an inoperable state in which the igniter switch is not configured to transmit or receive electrical signals, and an igniter ignitable by the igniter switch in response to the igniter switch receiving a firing signal addressed to the igniter switch, and configured to actuate the setting tool in response to ignition of the igniter when the setting tool initiator is connected to the setting tool, wherein the setting tool initiator is configured to shift the igniter switch from the operational state to the inoperable state in response to the ignition of the igniter. In some embodiments, the igniter switch comprises a digitally addressable switch. In some embodiments, the igniter switch comprises a processor and a memory device including instructions stored therein defining the operation of the igniter switch. In certain embodiments, the setting tool initiator comprises an initiator housing in which the igniter switch and the igniter are received, a switch chassis positioned in the initiator housing and having an interior in which the igniter switch is positioned, and an igniter adapter positioned in the initiator housing and coupled to the switch chassis, wherein the igniter adapter has an interior in which the igniter is positioned. In certain embodiments, the switch chassis positions the igniter switch at a predefined distance from the igniter. In some embodiments, the predefined distance is 1.75 inches or less. In some embodiments, the setting tool initiator further comprises an igniter spring positioned in the interior of the igniter adapter, wherein the igniter spring electrically connects the igniter switch with the igniter. In some embodiments, the setting tool initiator comprises an initiator housing in which the igniter switch and the igniter are received, and wherein the initiator housing defines an igniter compartment in which the igniter is positioned and a switch compartment in

which the igniter is positioned, wherein a fluid flowpath is formed extending through the initiator housing from the igniter compartment to the switch compartment. In some embodiments, the initiator housing has a maximum length of 6.5 inches or less. In some embodiments, the switch compartment has a maximum inner diameter of 1.50 inches or less.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the present disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.
- (2) FIG. 1 is a schematic illustration of an embodiment of a system for completing a subterranean well.
- (3) FIG. 2 is a cross-sectional view of an embodiment of a tool string for completing a subterranean well.
- (4) FIG. 3 is a partial cross-sectional view of a conventional setting tool initiator for activating a setting tool.
- (5) FIG. 4 is a partial cross-sectional view of an igniter switch assembly installed in a setting tool initiator according to an embodiment of the present disclosure.
- (6) FIG. 5 is a partial cross-sectional view of an igniter switch assembly according to an embodiment of the present disclosure.
- (7) FIG. 6 is a partial cross-sectional view of an igniter switch assembly according to another embodiment of the present disclosure.
- (8) FIG. 7 is a partial cross-sectional view of an igniter switch assembly according to still another embodiment of the present disclosure.
- (9) FIG. 8 is a cross-sectional view of a signal sub according to an embodiment of the present disclosure.
- (10) FIG. 9 is a partial cross-sectional view of an igniter switch assembly installed in a setting tool initiator according to another embodiment of the present disclosure.
- (11) FIG. 10 is a partial cross-sectional view of an igniter assembly installed in a setting tool initiator according to an embodiment of the present disclosure.
- (12) FIG. 11 is a partial cross-sectional view of an igniter switch assembly according to a further embodiment of the present disclosure.
- (13) FIG. 12 is a partial cross-sectional view of an igniter assembly installed in a setting tool initiator in a second position according to a further embodiment of the present disclosure.

DETAILED DESCRIPTION

(14) It should be understood at the outset that although illustrative implementations of one or more embodiments are illustrated below, the disclosed systems and methods may be implemented using any number of techniques, whether currently known or not yet in existence. The disclosure should in no way be limited to the illustrative implementations, drawings, and techniques illustrated below, but may be modified within the scope of the appended claims along with their full scope of equivalents.

(15) In the following discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection can be through a direct connection of the two devices, or through an indirect connection that is established via other devices, components, nodes, and connections. In addition, as used herein, the terms “axial” and “axially” generally mean along or parallel to a given axis (e.g., central axis of a body or a port),

while the terms “radial” and “radially” generally mean perpendicular to the given axis. For instance, an axial distance refers to a distance measured along or parallel to the axis, and a radial distance means a distance measured perpendicular to the axis. Any reference to up or down in the description and the claims is made for purposes of clarity, with “up”, “upper”, “upwardly”, “uphole”, or “upstream” meaning toward the surface of the borehole and with “down”, “lower”, “downwardly”, “downhole”, or “downstream” meaning toward the terminal end of the borehole, regardless of the borehole orientation.

(16) Tools used in completing oil well or gas wells are introduced or carried into a subterranean wellbore on a work string, such as wireline, electric line, continuous coiled tubing, threaded work string, or the like, for engagement at a pre-selected position within the wellbore. The wellbore can be lined with a tubular conduit such as a casing string or liner. The wellbore can be an openhole section where the drilled formation does not have the conduit supporting the drilled formation. The wellbore can include a secondary tubing member, such as production tubing, that is placed within a casing, liner, or openhole section. These completion tools include sealing devices such as expandable elastomeric plugs, permanent or retrievable plugs, bridge plugs, ball seats, packers, production packer, service packer, production sleeve, ball-type and other valves, injectors, perforating guns, tubing hanger, casing hanger, liner hanger, cement plug dropping heads, and other devices typically encountered during the drilling, completion, or remediation of a subterranean well. Such devices and tools will hereafter collectively be referred to as “auxiliary tools.” The auxiliary tool is typically set and anchored into position within the casing, tubing, or openhole section such that movements in various directions such as upwardly, downwardly, or rotationally, are resisted, and, in fact, prevented. Such movements can occur as a result of a number of causes, such as pressure differentials across the tool, temperature variances, tubing or other conduit manipulation subsequent to setting for activation of other tools in the well, and the like.

(17) The auxiliary tool typically must be set or actuated to position the auxiliary tool at the required depth within the casing, liner, tubing, or openhole section. In some cases, the auxiliary tool can comprise, for example, a plug or packer including a packing element that will form a seal when energized. As described above, the activation or manipulation of some of such auxiliary tools often is achieved by use of a setting tool which can be introduced into the wellbore along with or subsequent to the auxiliary tool on a work string, such as wire or electric line, continuous or coiled tubing, threaded tubing, drill pipe, or by other known means. In some applications, a setting tool can include one or more pistons to move or stroke a portion of the setting tool relative to stationary portion of the setting tool to apply a setting force in compression or in tension to the auxiliary tool. Pressure can be applied to face of the piston within the setting tool to generate the setting force to set or actuate the auxiliary tool.

(18) Some setting tools utilize an explosive or combustible element or power charge to develop a high pressure gas within a combustion compartment of the setting tool following the ignition of the combustible element. The high pressure generated by the burning or firing of the combustible element may drive a piston, stroking rod, or other member of the setting tool to move relative a stationary member to cause the manipulation of the auxiliary tool. By “burning” or “firing” it is meant the continuous generation, sometimes relatively slowly, of pressure by ignition of a combustible element initiated reaction which results in a pressure increase within a combustion compartment of transmittable gaseous pressure within the apparatus. Sometimes the terms “detonate” and “ignite” are used to describe a sudden generation of gaseous pressure. The terms “detonate”, “burning”, “igniting,” or “firing”, all describe the generation of gaseous pressure by the burning of the combustible element with different timescales.

(19) The ignition of the combustible element to burn is started with an igniter of a setting tool initiator or firing head coupled to the setting tool. The igniter can be comprised of a plurality of igniters. For example, the igniter can be a single primary igniter, a primary igniter and secondary igniter, or a primary, secondary, and embedded igniter. The primary igniter can comprise a tube, an

electronic ignition device, and a pyrotechnic material that creates a jet of heat and flame. Similarly, the secondary and embedded igniter can be comprised of a pyrotechnic material. The electronic ignition device within the primary igniter can be, for example, any combination of a thermal match, a heater cartridge, an electrical trigger, an electrical igniter, or an electrical arc generator. The pyrotechnic material can be any combination of gunpowder, thermite, or a metal and oxidizer mix. An electrical signal, e.g., a combination of voltage and current, applied to the primary igniter can ignite the pyrotechnic material with the electronic ignition device and produce a heated jet or flame jet extending outwards from the end of the tube as the pyrotechnic material is consumed. The flame jet from the igniter may contact and ignite the combustible element in the combustion compartment.

(20) The setting tool initiator including a setting tool housing can be installed onto the combustion compartment of the setting tool to house the igniter and to electrically communicate with the igniter. The setting tool initiator may also include an igniter switch electrically connected to both the igniter and to a controller located at the surface of the wellbore. In some applications, the igniter can be installed within the initiator housing and threadingly connected to the combustion compartment of the setting tool.

(21) In a typical deployment of a tool string including a conventional setting tool initiator, operator at the surface prepare the tool string for conveyance into the well. The tool string can comprise a work string, the setting tool initiator, a setting tool, and an auxiliary tool. The operator may releasably connect the auxiliary tool to the setting tool, install a combustible element into the combustion compartment of the setting tool, and connect the setting tool initiator igniter to the setting tool. The operator may then direct the conveyance of the tool string into the wellbore and convey the tool string through the wellbore to the desired location. The location of the tool string can be verified by any combination of a measured length of the tool string, the number of collars counted by a collar locator, a location device within the wellbore, or by a measurement transferred to surface by the setting tool initiator.

(22) Once at the desired location, the operator may signal the igniter switch of the setting tool initiator to ignite the igniter and activate the setting tool to set the auxiliary tool at the desired location. The operator at surface may not receive any indication that the igniter was initiated by the signal to the igniter switch. In one case, the operator may not know that the setting tool did not function. For example, undetected corrosion or the buildup of carbon around the ignition switch can prevent the signal or weaken the signal sent to the igniter. In another example, the operator may not know that the setting tool did function and set the auxiliary tool. In some shallow wells, the weight of the work string can indicate a change in value (e.g., the tool string weighs less, indicating that the setting tool did function). However, in deeper wells or on floating platforms offshore, there may not be a noticeable change in the weight of the work string. Generally, the work string cannot be lowered to bump or tag the auxiliary tool as the setting tool could become entangled with the set auxiliary tool in the well resulting in an expensive fishing job to retrieve the stuck tools string. Thus, it is desirable to develop a setting tool initiator that provides feedback to surface that the igniter did activate and ignited the combustible element within the setting tool.

(23) Embodiments described herein include a setting tool initiator that is comprising an igniter switch module is coupled to a combustion compartment of a setting tool. A switch module comprising the igniter switch and the igniter are installed into the setting tool initiator. An operator may signal the igniter switch to initiate the igniter. The igniter produces a flame jet that ignites the combustible element to burn. In this configuration, the igniter switch malfunctions due to the heat and pressure from the burning combustible element filling the setting tool initiator. The operator may register the malfunction igniter switch at surface as an indication that the setting tool has functioned to set the auxiliary tool.

(24) In an embodiment, the setting tool initiator is comprising an igniter switch module is coupled to the combustion compartment of the setting tool. The switch module comprises the igniter switch,

a circuit breaker, and the igniter. The circuit breaker can be a thermal switch, pressure switch, or an impact switch. The circuit breaker is electrically connected to the igniter switch and cuts off communication to the igniter switch when a predetermined threshold value is reached. For example, the thermal switch may break communication with the igniter switch when the temperature of the switch exceeds 500 degrees Fahrenheit (° F.). The switch module is installed into the setting tool initiator. The operator may signal the igniter switch to initiate the igniter. The igniter produces a flame jet that ignites the combustible element to burn. The circuit breaker disconnects the igniter switch due to the heat and pressure exceeding a preset value of the circuit breaker from the burning combustible element filling the setting tool initiator. The operator may register the end of communication with the igniter switch at surface as an indication that the setting tool has functioned to set the auxiliary tool.

(25) In an embodiment, the setting tool initiator is comprising an igniter switch module is coupled to the combustion compartment of the setting tool. The switch module comprises the igniter switch, an instrument sensor, and the igniter. The instrument sensor can be an electronic temperature sensor, pressure sensor, an impact sensor, or an acoustic sensor. The instrument sensor communicates data through the switch module to surface via the conductors in the work string. The instrument sensor can communicate data continuously after the signal is sent to the igniter switch. For example, the temperature sensor communicates temperature measurements from inside the setting tool initiator to the operator at surface. In operation, the operator may signal the igniter switch to initiate the igniter. The igniter produces a flame jet that ignites the combustible element to burn. The temperature sensor communicates the temperature measurements to surface via the electrical conductor in the work string. The service personnel monitor the temperature of the setting tool initiator from the surface. An increase in the temperature of the setting tool initiator indicates that the setting tool has functioned to set the auxiliary tool.

(26) The present disclosure describes a setting tool apparatus for use in a wellbore comprising a switch module capable of communicating the switch status to surface after initiating the igniter to power the setting tool to activate a downhole completion tool. The communication by the switch module to the surface personnel indicates the status of the setting tool saving the service personnel time and resources diagnosing the downhole status of the setting tool and downhole completion tool.

(27) Referring now to FIGS. **1** and **2**, an embodiment of a system **10** for plugging a wellbore **14** extending from the surface **5** through a subterranean earthen formation **16** is shown. In this exemplary embodiment, plugging system **10** generally includes a surface assembly or servicing rig **12** positioned at the surface **5** that extends over and around the wellbore **14** that penetrates the earthen formation **16** for the purpose of recovering hydrocarbons from a first production zone **18A** and a second production zone **18B** (collectively the production zones “**18**”). The wellbore **14** can be drilled into the subterranean formation **16** using any suitable drilling technique. While shown as extending vertically from the surface in FIG. **1**, the wellbore **14** can also be deviated, horizontal, and/or curved over at least some portions of the wellbore **14**. For example, the wellbore **14**, or a lateral wellbore drilled off of the wellbore **14**, may deviate and remain within one of the production zones **18**. The wellbore **14** can be cased, open hole, contain tubing, and can generally be made up of a hole in the ground having a variety of shapes and/or geometries as is known to those of skill in the art. In the illustrated embodiment, a casing **20** can be placed in the wellbore **14** and secured at least in part by cement **22**.

(28) The servicing rig **12** of plugging system **10** can be one of a drilling rig, a completion rig, a workover rig, a wireline surface system, or other structure and supports a tool string **32** disposed in the wellbore **14**. Servicing rig **12** includes a surface controller **13** in signal communication with one or more downhole tools of tool string **32**. In other embodiments, other surface systems or structures can also support the tool string **32**. The servicing rig **12** can also comprise a derrick with a rig floor through which the tool string **32** extends downward from the servicing rig **12** into the wellbore **14**.

It is understood that other mechanical mechanisms, not shown, can control the run-in and withdrawal of the tool string **32** in the wellbore **14**.

(29) In this exemplary embodiment, the tool string **32** generally includes a work string **30**, a perforating gun **46** (hidden from view in FIG. 2), a signal sub **34**, a setting tool initiator, a setting tool **42**, and an auxiliary tool **44**. It may be understood that in other embodiments the configuration of tool string **32** may vary. For example, in some embodiments, tool string **32** may additionally include a fishneck, one or more weight bars, a release tool, and/or one or more other downhole tools. The work string **30** can be any of a string of jointed pipes, a slickline, a coiled tubing, and a wireline. The auxiliary tool **44** may comprise one or more frac plugs, one or more packers, one or more tubing hangers, one or more completion components such as screens and/or production valves, sensing and/or measuring equipment, and other equipment which are not shown in FIG. 1. The tool string **32** can be lowered into the wellbore **14** to position the setting tool **42** to set or actuate a frac plug at a predetermined depth.

(30) As shown particularly in FIG. 2, in this exemplary embodiment, setting tool **42** generally includes a setting tool housing **43**, a piston **48** slidably disposed in the housing **43**, and a combustible or explosive element **49** positioned in the setting tool housing **43**. Particularly, setting tool housing **43** defines a central passage **51** having a combustion compartment **53** in which the combustible element **49** is received. Piston **48** is configured to impart a setting force against the auxiliary tool **44** in response to combustion of the combustible element **49**. While the setting tool initiator **40** is described herein as separate from the setting tool **42**, it may be understood that in some embodiments the setting tool initiator **40** may comprise a component of the setting tool **42** with the initiator housing comprising a section (e.g., a section housing) of the setting tool housing **43**.

(31) Auxiliary tool **44** is releasably attached to a distal or downhole end of the setting tool **42**. In this exemplary embodiment, the signal sub **34** includes any combination of a cable head **36**, and an instrument sub **38**. The cable head **36** attaches the signal sub **34** to a work string **30** that includes an electrical conductor **28**. For example, a wireline can include one or more electrical conductors wrapped with a braided wire. The cable head **36** can electrically connect the one or more electrical conductors **28** to another component of the signal sub **34** as will be described herein. The perforating gun includes one or more explosive shaped charges configured to perforate casing **20** at the desired location in response to receiving, by a gun switch of the perforating gun, a firing signal from the surface controller **13**. It may be understood that while only a single perforating gun **46** is shown in FIG. 1, in other embodiments, tool string **32** may include more than one perforating gun **46**.

(32) In this exemplary embodiment, signal sub **34** of tool string **32** includes an instrument sub **38** with environmental sensors **56**. The instrument sub **38** couples to the cable head **36** with an electrical connection **54**. The environmental sensors **56** can include pressure and temperature sensors to measure the pressure and temperature of the wellbore environment, the pressure and temperature of the interior of the instrument sub, or a combination of both. The environmental sensor **56** can include a motion sensor that can be one or more accelerometers. The measurements of the accelerometers can indicate motion of the setting tool **42**. The environmental sensor **56** can include a magnetic sensor commonly referred to as a collar locator used to indicate the location of the setting tool initiator within the wellbore **14**. In some embodiments, the environmental sensor **56**, of instrument sub **38** may only comprise the magnetic sensor. In some embodiments, other components of the tool string **32** such as perforating gun **46** may be positioned between the instrument sub **38** and setting tool **42**.

(33) The setting tool initiator **40** may connect to the signal sub **34** with an electrical connector sub **60** configured to provide a sealed electrical connection between the setting tool initiator **40** and the signal sub **34**. The upper sealed electrical connection **60** electrically couples the setting tool initiator **40** to the electrical conductors **28** in the work string **30**. The upper sealed electrical

connection **60** can also provide pressure isolation between the setting tool initiator **40** and components of tool string **32** positioned uphole from setting tool initiator **40** such as, for example, perforating gun **46**.

(34) Turning now to FIG. **3**, a conventional setting tool initiator **400** is shown. Setting tool initiator **400** generally includes an initiator housing **402**, an igniter **420**, and a setting tool igniter switch **425**. Initiator housing **402** is shown as including a pair of housing sections **403** and **405** which are connected together to form initiator housing **402**. However, it may be understood that initiator housing **402** may comprise only a single housing or more than two housings. Initiator housing **402** defines an internal igniter compartment **404** and an internal switch compartment **406** within the housing **402**. The igniter **420** is located in the igniter compartment **404** while the igniter switch **425** is located in the switch compartment **406**. The igniter switch **425** is electrically connected to the igniter **420** via an electrical connector located in the initiator housing **402**. In this manner, igniter switch **425** may transmit an electrical signal to the igniter **420** through the electrical connector **440** to ignite the igniter **420**. The setting tool initiator **400** may thus activate setting tool **42** (not shown in FIG. **3**) in response to the ignition of igniter **420**.

(35) Conventionally, the igniter **420** is separate from the igniter switch **425** by a bulkhead **415** positioned within initiator housing **402** between the igniter **420** and igniter switch **425**. The bulkhead **415** may be separate from or integrated with the electrical connector **410**. Conventionally, the bulkhead seals and provides a pressure barrier between the switch compartment **406** and the igniter compartment **404** such that hot and highly pressurized combustion gasses produced by the ignition of igniter **420** are prevented from entering the switch compartment **406** and thereby physically compromising or disabling the igniter switch **425**. In this manner, the igniter switch **425** may remain in signal communication with the surface controller **13** following the ignition of igniter **420**. For instance, the igniter switch **425** may be used to perform additional actions such as detonating the one or more shaped charges of the perforating gun **46** following the ignition of igniter **420**.

(36) While the conventional setting tool initiator **400** is configured to permit igniter switch **425** to survive the ignition of igniter **420**, the survival of igniter switch **425** in-turn prevents the destruction or disablement of igniter switch **425** from providing a surface indication to the operator of system **10** that the setting tool **42** has successfully been activated to set the auxiliary tool **44**. Instead, the operator at the surface is forced to rely on more time consuming (and hence costly) and less reliable techniques for deciphering whether the auxiliary tool **44** has been successfully set, such as by applying tension to the work string **30** using the servicing rig **12** to determine if the auxiliary tool **44** has anchored against the casing **20**. However, as described above, in some applications (e.g., relatively deep wells, off-shore applications) it is difficult if not impossible to determine whether the auxiliary tool **44** has been successfully set based on tension applied to the work string **30** as observed at the surface.

(37) It may also be understood that if bulkhead **415** were removed from the conventional setting tool initiator **400** to intentionally compromise igniter switch **425** following the ignition of igniter **420**, such a modification would require the combustion products produced by the combustible element of setting tool **42** to fill both the igniter compartment **404** and switch compartment **406**. However, the igniter switch **425** is not positioned proximal igniter **420**, and the switch compartment **406** has a relatively large volume compared to the volume of igniter compartment **404**. The large volume of switch compartment **406**, when filled with combustion products produced by the combustible element of setting tool **42**, reduces the pressure force imparted by the combustion products against the piston **48** of setting tool **42**, concomitantly reducing the setting force applied by the piston **48** of setting tool **42** to the auxiliary tool **44** for setting or actuating the auxiliary tool **44**. Particularly, the increased volume occupied by the combustion products in the switch compartment **406** reduces the pressure of the combustion products by increasing the volume the products are permitted to expand into, reducing the effectiveness of the setting tool **42** in setting

the auxiliary tool **44** by reducing the pressure force exerted by the setting tool **42** during actuation. (38) Turning now to FIG. **4**, an embodiment according to the current disclosure of the setting tool initiator **40** is shown. As will be explored in further detail below, unlike conventional setting tool initiator **400** shown in FIG. **3**, setting tool initiator **40** of the current disclosure is configured to provide a surface indication of the successful ignition of an igniter **130** of the setting tool initiator **40** by disabling or disconnecting an electrical igniter switch **110** of the setting tool initiator **40**. In this exemplary embodiment, setting tool initiator **40** generally includes an initiator housing **74** and an igniter switch module **70**. The setting tool initiator **40** may connect with uphole components of tool string **32** (e.g., cable head **36**) via the connector sub **60** shown in FIG. **2** and hidden from view in FIG. **4**. As will be described further herein, igniter switch module **70** is configured to place igniter switch **110** in close proximity with igniter **130** whereby combustion products may be communicated to the igniter switch **110** while minimizing the amount of additional volume the combustion products must occupy following the ignition of igniter **130**. In this manner, igniter switch module **70** permits the compromising of igniter switch **110** to serve as a surface indication of the successful actuation of setting tool **42** while also maximizing the effectiveness of setting tool **42** (by maximizing the pressure force exerted by setting tool **42** during actuation) in setting or actuating the auxiliary tool **44**.

(39) In this exemplary embodiment, the initiator housing **74** is a cylindrical shape with an uphole connector **78**, a downhole connector **80**, and a central bore or passage **81** extending between longitudinally opposed uphole and downhole ends of the initiator housing **74**. In this exemplary embodiment, initiator housing **74** comprises a single, integrally or monolithically formed housing and the central passage **81** thereof receives the entirety of the igniter switch module **70**. It may be understood however that in other embodiments initiator housing **74** may comprise a plurality of separate sectional housings which are threaded or otherwise connected together end-to-end.

(40) In this exemplary embodiment, central passage **81** of initiator housing **74** includes a switch compartment **82**, and an igniter compartment **86** that is connected to the switch compartment **82** by an unabridged interrupt flowpath **85** extending from the igniter compartment **86** to the switch compartment **82**. In some embodiments, the interrupt flowpath **85** extends from the combustion compartment **53** and to the switch compartment **82** such that combustion products may be conveyed from the combustion compartment **53** to the switch compartment **82**. The switch compartment **82** has an inner housing surface **98**, a grounding surface **88**, and transitions to the igniter compartment **86**. The uphole connector **78** includes an upper seal surface **92** to seal against a corresponding seal assembly of the connector sub **60** to prevent well bore fluids from entering the initiator housing **74**. The downhole connector **80** includes a seal assembly **96** configured to seal against a corresponding seal surface defining the combustion compartment **53** of the setting tool **42**. The igniter switch module **70** can be installed inside the switch compartment **82** of the initiator housing **74**. The igniter attached to the igniter switch module **70** installs into the igniter compartment **86**. Initiator housing **74** is configured to minimize the volume of switch compartment **82** such that the volume occupied by the combustion products generated by setting tool **42** during actuation is low enough such that the combustion products may maintain a pressure sufficient to fully set or actuate the auxiliary tool **44**. In this exemplary embodiment, the switch compartment **82** has a maximum inner diameter of 1.50 inches (in) or less to thereby minimize the volume of switch compartment **82**; however, it may be understood that the maximum inner diameter of switch compartment **82** may vary in other embodiments.

(41) The igniter switch module **70** can be tested by the operator for electric connectivity before being installed into the switch compartment **82**. As an example, the operator may measure electrical resistance of the igniter **130** after being installed into the igniter switch module **70** by contacting a first lead of a resistance meter to downhole electrical contact **120** and contacting a second lead of the meter to tube **132**. Turning now to FIG. **5**, in this exemplary embodiment, the igniter switch module **70** generally includes a main body or switch chassis **112**, igniter switch **110**, an igniter

adapter **140**, and igniter **130**. Igniter switch module **70** allows for the igniter switch **110** and igniter **130** to be pre-connected and installed together as a single unit into the initiator housing **74**. As described above, igniter switch module **70** places the igniter switch **110** into close proximity with the igniter **130** so as to maximize the effectiveness of setting tool **42** during actuation. The igniter switch module **70** has a maximum length **111** extending from an uphole end of the switch chassis **112** to a downhole end of the igniter adapter **140**. In this exemplary embodiment, the maximum length **111** of igniter switch module **70** is approximately 6.5 in or less; however, it may be understood that the maximum length **111** of igniter switch module **70** may vary in other embodiments.

(42) The switch chassis **112** of igniter switch module **70** may be made of a non-electrically conductive material (e.g., plastic) such as glass filled nylon. Switch chassis **112** has an uphole electrical contact **114** and a downhole electrical contact **120** for communicating signals to the igniter **130** as will be disclosed further herein. In this exemplary embodiment, igniter adapter **140** includes a tube **132**, a flange **124**, and a ground or flange spring **136**. The tube **132** may be connected or attached to a flange **124** by a weld **138**, by fasteners, or by other means. Flange spring **136** may be connected or attached to the flange **124** by a weld **138**, by a bent tab, by fasteners, or by other means.

(43) In this exemplary embodiment, igniter switch module **70** additionally includes an igniter spring **128** and a shoulder washer **126**. Igniter spring **128** and shoulder washer **126** are installed between the switch chassis **112** and the igniter adapter **140**. Tube **132** comprises one or more tabs that bend outwards to secure the tube **132** to the switch chassis **112** and to secure the flange spring **136**. The igniter adapter **140** may be attached to the switch chassis **112** with fasteners such as screws. In this exemplary embodiment, igniter **130** is installed into the tube **132** of the igniter adapter **140** and secured in place with a snap ring **134** or any other suitable fastener. Igniter switch **110** is connected to the uphole electrical contact **114** with an uphole switch wire **116**. Additionally, igniter switch **110** is connected to the downhole electrical contact **120** with a downhole switch wire **118**. A grounding wire **122** from the igniter switch **110** may be connected to a screw or similar location on the front of the igniter adapter **140**. The uphole switch wire **116**, downhole switch wire **118**, and igniter switch **110** collectively form a switch circuit **115** (shown in FIG. 5) which is electrically disconnected in response to the circulation of combustion products to the switch compartment **82** and the concomitant exposure of the switch circuit **115** to the combustion products. For example, one or more of the wires **116** and **118** and igniter switch **110** may be physically compromised following circulation of the combustion products to the switch compartment **82**. Additionally, while in this exemplary embodiment the igniter switch **110** is positioned in the switch compartment **82**, in other embodiments, igniter switch **110** may be positioned external the switch compartment **82** with another portion of the switch circuit **115** (e.g., downhole switch wire **118**) positioned in the switch compartment **82**.

(44) The igniter switch **110** has an operational state in which the igniter switch **110** is configured to receive electrical signals from the surface **5** and an inoperable state in which the igniter switch **110** is not configured to receive electrical signals from the surface **5**. Setting tool initiator **40** is configured to shift igniter switch **110** from the operational state to the inoperable state in response to the ignition of the igniter **130** which results in the communication of combustion products to the switch chamber **82**. For example, the igniter switch **110** may be shifted to the inoperable state by rendering electrically inoperable (e.g., physically compromising) the igniter switch **110** itself or another component of the switch circuit **115** such as uphole switch wire **116**.

(45) Igniter switch module **70** positions the igniter switch **110** at a predefined distance **113** from the igniter **130**, where the predefined distance is contingent or based on the length of the switch chassis **112**, and the length of igniter switch **128** when compressed by the igniter **130**. It may be understood that a limited degree of movement may be permitted between igniter switch **110** and igniter **130** and thus the predefined distance **113** may comprise a predefined range. For example, in some

embodiments, the predefined distance **113** is approximately 1.75 in or less; however, it may be understood that in other embodiments the predefined distance **113** may vary.

(46) Signals transmitted from an operator at the surface can be communicated to the igniter **130** as will be described herein. For example, the operator may transmit an igniter signal down the electrical conductor **28** within the work string **30** to the tool string **32** shown in FIG. 2. The igniter signal is communicated from the electrical conductor **28** within the work string **30**, through the electrical contacts within the signal sub **34**, and to the setting tool initiator **40** shown in FIG. 4 via the connector sub **60**. From connector sub **60**, the igniter signal travels to the igniter switch module **70**. The transmitted signal passes through the uphole contact **114** and, the uphole switch wire **116**, and to the igniter switch **110**. In some embodiments, the igniter switch **110** comprises an addressable switch, including, for example, a printed circuit board, a processor (e.g., a microprocessor or central processing unit (CPU)), and a memory device including instructions stored therein defining the operation of igniter switch **110**. The igniter switch **110** has an operational state or configuration in which the igniter switch can receive signals transmitted from surface. For example, when in the operational state, igniter switch **110** may identify an address and a command within the signal, compare the transmitted address to the programmed address within the memory of the igniter switch **110**, and execute the command if the transmitted address matches the address in memory. If the transmitted address matches the address in memory, a firing circuit of the igniter switch **110** is opened and permits the voltage and current to be provided to the igniter **130** via the downhole switch wire **118**, the downhole contact **120**, and the igniter spring **128**. As will be discussed further herein, igniter switch **110** additionally includes a disabled or compromised state or configuration in which the switch **110** is not configured to receive signals transmitted from the surface. For example, in the disabled state the igniter switch **110** may be damaged or otherwise physically compromised. As another example, in the disabled state, the circuit connecting igniter switch **110** to the surface controller **13** may be physically damaged or otherwise compromised. It may also be understood that in other embodiments the configuration of igniter switch **110** may vary. For example, in other embodiments, igniter switch **110** may comprise a diode-based switch and may not include a processor or a memory device.

(47) The igniter **130** is grounded to the igniter adapter **140** via biasing members or springs integral to the body of the igniter **130** that contact the inner surface **142** of the tube **132** of the igniter adapter **140**. The igniter adapter **140** is grounded to initiator housing **74** of the setting tool initiator **40**, as shown in FIG. 4, via the flange spring **136** in contact with the grounding surface **88** of the initiator housing **74**. The igniter switch **110** may also be grounded to the grounding surface **88** of the initiator housing **74** via grounding wire **122** that is connected to the igniter adapter **140**.

(48) The igniter **130** ignites in response to the igniter switch **110** conveying the signal (e.g., the necessary voltage and current) necessary to initiate the pyrotechnic material of the igniter **130**. The resultant flame jets out of the downhole end of the igniter **130** to ignite the combustible element **49** within the combustion compartment **53** of the setting tool **42**. The burning or detonation of the combustible element **49** creates a high pressure and high temperature gaseous pressure within the combustion compartment **53** that strokes the piston **48** of the setting tool **42** to set or actuate the auxiliary tool **44**. The high pressure and high temperature gases pass between the outer surface **74** of the tube **132** on the igniter adapter **140** and the inner surface **90** of the igniter compartment **86** of the initiator housing **74** to fill the switch compartment **82** of the setting tool initiator **40**. In this manner, the environment within the switch compartment **82** of the setting tool initiator **40** changes from a pressure near atmospheric pressure (e.g., 14.7 psi) to a substantially elevated pressure (e.g., a pressure exceeding 10,000 pounds per square inch (PSI)).

(49) As a result of ignition, the igniter switch **110** short circuits, e.g., creates an open circuit, due the change in environmental conditions within the switch compartment **82**, e.g., high pressure and high temperature of the gases within the switch compartment **82**. Hot pressurized combustion products generated by the ignition of igniter **130** and of the combustible element **49** of the setting

tool **42** (the combustible element **49** being in fluid communication with igniter **130**) are communicated or flow along flowpath **85** shown in FIG. **4** from the igniter compartment **86** to the switch compartment **82** where the combustion products contact the igniter switch **110** and shift the igniter switch **110** from the operational state to the disabled state. Particularly, the combustion products physically damage or otherwise compromise the physical integrity of igniter switch **110** and/or other circuitry connected thereto (e.g., uphole switch wire **116**) whereby igniter switch **110** is no longer connected to surface controller **13** or configured to send or receive signals.

(50) The operator at surface may register the short circuit, i.e., end of communication, as a positive and mechanical surface indication that the combustible element **49** within the setting tool **42** has burned and actuated the setting tool **42** to activate the auxiliary tool **44**. In this manner, the operator need not rely on the unreliable practice of applying tension to work string **30** at the surface to determine whether the auxiliary tool **44** has been set. Moreover, igniter switch module **70** places the combustible element **49** and particularly igniter **130** into close proximity with igniter switch **110**, thereby ensuring the destruction of igniter switch **110** while minimizing the volume of the central passage **81** of initiator housing **74** and thus the volume which is occupied by the combustion products following the ignition of the igniter **130**. Minimizing the volume occupied by the combustion products generated by the ignition of igniter **130** and the combustible element **49** maximizes the pressure force imparted by the combustion products to the piston **48** of the setting tool **42** which strokes in response to the ignition of the igniter **130**. The minimization of the volume of central passage **81** may thus assist in ensuring the piston **48** of setting tool **42** fully strokes to thereby fully and successfully set the auxiliary tool **44**.

(51) In an embodiment, a circuit breaker in the igniter switch module **70** disconnects the communication path to the igniter switch **110**. Turning now to FIG. **6**, in this embodiment, an igniter switch module **80** comprises the igniter switch **110**, a main body **154** housing the igniter switch **110**, a circuit breaker **156**, the igniter adapter **140**, and the igniter **130**. The circuit breaker **156** can be a thermal switch, pressure switch, or an impact switch. The circuit breaker **156** is electrically connected within the circuit between the uphole contact **114** and the igniter switch **110**. An electronic signal transmitted from surface controller **13** is communicated through the electrical conductor **28** in the work string **30**, through the signal sub **34**, and to the uphole contact **114** on the igniter switch module **80**. In this exemplary embodiment, the signal from surface controller **13** passes through the uphole contact, a second switch wire **82**, the circuit breaker **156**, the uphole switch wire **88**, to the igniter switch **110**. The electronic signal from surface controller **13** may pass through the circuit breaker **156** until a predetermined value is reached and the circuit breaker **156** cuts off communication to the igniter switch. If the circuit breaker **156** is a thermal switch, the thermal switch breaks communication with the igniter switch **110** when the temperature exceeds a predetermined value (e.g., 500 degrees Fahrenheit (° F.)). If the circuit breaker **156** is an impact switch, the impact switch (i.e., accelerometer) breaks communication with the igniter switch **110** when the impact force (i.e., acceleration) exceeds a predetermined value (e.g., 10 g).

(52) In this exemplary embodiment, when the surface controller **13** transmits an electronic signal to the igniter switch **110** and the transmitted address matches the address in memory, the igniter switch **110** opens the firing circuit thereof to permit the transmission of the voltage and current to the igniter **130** via the downhole switch wire **118**, the downhole contact **120**, and the igniter spring **128**. The igniter **130** ignites and the resultant flame jets out to ignite the combustible element **49** within the combustion compartment **53** of the setting tool **42**. The burning or detonation of the combustible element **49** creates a high pressure and high temperature gaseous pressure within the combustion compartment **53** that strokes the piston **48** on the setting tool **42** to set or actuate the auxiliary tool **44**. The high pressure and high temperature gases pass between the outer surface **74** of the tube **132** on the igniter adapter **140** and the inner surface **90** of the igniter compartment **86** of the initiator housing **74** to fill the switch compartment **82** of the setting tool initiator **40**. The circuit breaker **156** disconnects or breaks communication with the igniter switch **110** when a

predetermined value is reached or exceeded. For example, if the circuit breaker **156** is a pressure switch, the pressure switch breaks communication with the igniter switch **110** when the pressure exceeds a predetermined value (e.g., 10,000 PSI). The operator may register the end of communication, or a break in communication, with the igniter switch **110** at surface controller **13** as an indication that the setting tool **42** has functioned to set the auxiliary tool **44**.

(53) In an embodiment, an environmental sensor within the switch module indicates the setting tool **42** has functioned. Turning to FIG. 7, in this embodiment, an igniter switch module **160** comprises the igniter switch **166**, an environmental sensor **162**, the igniter adapter **140**, and the igniter **130**. The environmental sensor **162** can be a thermometer, a pressure transducer, an accelerometer, or an acoustic sensor. The igniter switch module **160** can have any combination of one or more environment sensors **162**. The environmental sensors **162** are electrically connected to the igniter switch **166** with a sensor wire **164**. In this exemplary embodiment, an electronic signal transmitted from surface controller **13** is communicated through the electrical conductor **28** in the work string **30**, through the signal sub **34**, and to the uphole contact **114** on the igniter switch module **160**. The signal transmitted from surface controller **13** passes through the uphole contact **114**, the uphole switch wire **116**, to the igniter switch **166**. As previously described, the igniter switch **166** can be an addressable switch. Likewise, the one or more environmental sensors **162** can be addressable through the addressable igniter switch **166**.

(54) An electronic signal from surface controller **13** can command the igniter switch **166** to transmit one or more measurements at a predetermined periodic rate from the environmental sensors **162**. For example, the environmental sensor **162** can be a temperature sensor (e.g., thermocouple) that measures the temperature within the switch compartment **82** of the initiator housing **74**. For example, the environmental sensor **162** can be a pressure sensor (e.g., pressure transducer) that measures the pressure within the switch compartment **82** of the initiator housing **74**. As another example, the environmental sensor **162** can be an accelerometer that measures the acceleration (e.g., motion) of the initiator housing **74**. As another example, the environmental sensor **162** can be an acoustic sensor (e.g., microphone, piezoelectric transducer) that measures the acoustic waves or sound levels within the switch compartment **82** of the initiator housing **74**. The surface controller **13** may transmit an electronic signal with a command to activate to the igniter **130** and a second command to transmit the measurements at a predetermined periodic rate from the environmental sensor **162**.

(55) When the igniter switch **110** receives the commands, the igniter switch **110** transmits a signal (e.g., a predetermined voltage and current) to the igniter **130** via the downhole switch wire **118**, the downhole contact **120**, and the igniter spring **128**. The igniter switch **166** can measure and transmit the measured data from the one or more environmental sensors **162**. The igniter **130** ignites and the resultant flame jets out the distal end to ignite the combustible element **49** within the combustion compartment **53** of the setting tool **42**. The burning or detonation of the combustible element **49** creates a high pressure and high temperature gaseous pressure within the combustion compartment **53** that strokes the piston **48** of the setting tool **42** to set or actuate the auxiliary tool **44**. The service personnel receive the transmitted data from the one or more environmental sensors **162**. The change of measured data, for example an increase in the temperature, observed at surface can indicate that the setting tool **42** has functioned to set the auxiliary tool **44**.

(56) In an embodiment, the signal sub **34** has a plurality of environmental sensors in two or more locations that provide feedback to the operator at the surface that the setting tool **42** has functioned to set or activate an auxiliary tool **44**. The setting tool initiator **40** can include the igniter switch module **160** with one or more environmental sensors **162**. The instrument sub **38** can include one or more environmental sensors **56**. The environmental sensors can have an internal sensor **172**, an external sensor **174**, or any combination thereof. The internal sensor **172** can provide measurements at a predetermined periodic rate of the environment inside the instrument compartment **176**. The external sensor **174** can provide measurements at a predetermined periodic

rate of the wellbore environment exterior of the instrument sub **38**. The environmental sensor **56** can be one or more of a temperature sensor, a pressure transducer, an accelerometer, a magnetic sensor, or an acoustic sensor. The environmental sensor **56** can include pressure and temperature sensors to measure the pressure and temperature of the wellbore environment, the pressure and temperature of the instrument compartment **176** of the instrument sub **38**, or any combination thereof. The environmental sensor **56** can include a motion sensor that can be one or more accelerometers. The measurements of the accelerometers can indicate motion of the setting tool. The environmental sensor **56** can include a magnetic sensor commonly referred to as a collar locator. The magnetic sensor measures the magnetic response of the casing, liner, or tubing. The collars that connect the casing, liner, or tubing have a different magnetic signature than the tubing bodies. The collar locator measures and counts the collars. The number of collars counted can be correlated to a tubing tally to indicate the location of the setting tool initiator within the wellbore. The environmental sensor **56** can include an acoustic sensor (e.g., microphone, piezoelectric transducer) that measures the acoustic waves or sound levels within the instrument compartment **176** of the instrument sub **38** or the acoustic waves external to the instrument sub **38**.

(57) As previously described, the surface controller **13** transmit a signal to the igniter switch module **160** to ignite the igniter **130** and subsequently ignite the combustible element **49** in the setting tool **42**. The surface controller **13** can also transmit a signal to the environmental sensor **162** on the igniter switch module **160** and the environmental sensor **56** within the instrument sub **38**. The environmental sensor **162** and environmental sensor **56** can measure at a predetermined periodic rate and transmit the measurements to service personnel at surface. Any combination of measured data from the instrument sub **38** or the igniter switch module **160** observed at surface by the operator can indicate the that the setting tool **42** has set the auxiliary tool **44**. For example, an increase in the temperature measured by the environmental sensor **162** within the igniter switch module **160** along with motion measured by the environmental sensor **56** within the instrument sub **38** can indicate that the setting tool **42** has functioned to set the auxiliary tool **44**.

(58) In an embodiment, the signal sub **34** can comprise an instrument sub **38** with one or more environmental sensors **56**, and the setting tool initiator **40** may include circuit breaker **156**. As previously described, the surface controller **13** can transmit a signal to the igniter switch module **160** to ignite the igniter **130** and subsequently ignite the combustible element **49** in the setting tool **42**. The service personnel can also transmit a signal to the environmental sensor **56** within the instrument sub **38**. The environmental sensor **56** can measure at a predetermined periodic rate and transmit the measurements to the operator at surface. The operator can monitor communication with the igniter switch module **160** within the setting tool initiator **40**. The circuit breaker **156** will end electrical communication with the igniter switch module **160** when a predetermined environmental condition is met. Any combination of measured data from the instrument sub **38** or loss of electrical communication with the igniter switch module **160** observed at surface by the operator can indicate the that the setting tool **42** has set the auxiliary tool **44**.

(59) The pressure within the combustion compartment **53** of the setting tool **42** after the combustible element **49** is ignited can actuate a piston **48** to ground out the igniter switch assembly. In an embodiment shown in FIG. **9**, the setting tool initiator **200** includes a movable isolator that grounds out the igniter switch assembly. In this exemplary embodiment, the setting tool initiator **200** generally includes a switch housing **202**, an igniter retainer **204**, a movable isolator **208**, and an igniter switch module **270**. The switch housing **202** is a cylindrical shape with an uphole connector **78**, a downhole connector **80**, an inner thread **212**, a switch compartment **220**, and an igniter compartment **222** connected to the switch compartment **220** by an uninterrupted fluid flowpath. In this exemplary embodiment, the switch compartment **220** has an inner housing surface **224**, and an isolator port **226**. The uphole connector **78** includes an upper seal surface **92**. The downhole connector **80** includes a lower seal assembly **96**. The housing connector **72** sealingly couples to the switch housing **202** to form a seal to prevent well bore fluids from entering the switch compartment

220. The downhole connector **80** and seal assembly couple the setting tool initiator **200** to the combustion compartment **53** of the setting tool **42**. The installation of the igniter switch module **270** and the igniter will be explained in more detail herein.

(60) Turning to FIG. **10**, the igniter assembly **240** can be installed into the igniter compartment **222**. In this exemplary embodiment, the igniter assembly **240** generally include an insulated pin connector **242**, a retaining spring **244**, movable isolator **208**, and an igniter **246**. The insulated pin connector **242** and movable isolator have an electrically conductive core to communicate electrical signals to the igniter **246**. The insulated pin connector **242** has an outer shell of insulating material. The movable isolator has a seal assembly **248** that can comprise one or more seals with various seal retaining structures. The igniter **246** includes a grounding spring **250** that electrically couples to the igniter compartment **222** of the switch housing **202**. The insulated pin connector **242** is coupled to the movable isolator **208** by threads, fasteners, welding, or similar joining methods. The retaining spring **244** can be installed over the insulated pin connector **242** and movable isolator **208**. The retaining spring **244**, insulated pin connector **242**, and movable isolator **208** with seal assembly **248** can be installed into the igniter compartment **222**. The igniter **246** can be installed into the igniter compartment **222** and retained with the igniter retainer **204**.

(61) The igniter switch module can be tested by the operator for electric conductivity before being installed into the setting tool initiator. Turning now to FIG. **11**, the igniter switch module **270** can comprise, a main body **272**, an igniter switch **274**, an upper pin assembly **276**, a lower pin assembly **278**, and a grounding point assembly. The main body **272** can be made of a non-electrically conductive material (e.g., plastic) such as a glass filled nylon. The upper pin assembly **276** comprises a pin connector **300**, a connector post **280**, a connector spring **282**, and a spring retainer **284**. The connector spring **282** and spring retainer **284** slidably fit over the connector post **280** with an allowance fit. The pin connector **300** can couple to the connector post **280** with threads, fasteners, or any other method of joining. The upper pin assembly **276** can threadably connect to the main body **272** with a threaded connection **310**. In this exemplary embodiment, the lower pin assembly **278** comprises a pin connector **288**, a connector post **290**, a connector spring **292**, and a spring retainer **294**. The lower pin assembly **278** can threadably connect to the main body **272** with a thread connection **296**. In this exemplary embodiment, the igniter switch module **270** includes a grounding point assembly **314** comprising a washer **316**, a fastener **318**, and a grounding wire connector **320**. The fastener **318** can thread into a port **308** to attach the grounding point assembly **314** onto the main body **272**. The igniter switch **274** can be connected to the upper pin assembly **276** with an uphole switch wire **302** and connected to the lower pin assembly **278** with a downhole switch wire **304**. A grounding wire **306** from the igniter switch **274** can be connected to the grounding wire connector **320** of the grounding point assembly **314**.

(62) The pressure inside the setting tool **42** will ground out the igniter switch **274**. Returning to FIG. **9**, the setting tool initiator is assembled by installing the igniter assembly **240** into the igniter compartment **222** and threadably connecting the igniter retainer **204** to the switch housing **202**. The igniter switch module **270** can be tested before installing into the switch compartment **220** of the switch housing **202**. The housing connector **72** is threadably connected to the switch housing **202**. The switch housing **202** is threadably coupled to the setting tool **42** with the downhole connector **80** and seal assembly **96** of switch housing **202**. Turning to FIG. **10**, the retaining spring **244** bias the movable isolator **208** towards the isolator port **226**. The igniter **246** is pushed into contact with the igniter retainer **204** by the spring force of the retaining spring **244**. The atmospheric pressure on either side of the seal assembly **248** on the movable isolator **208** is approximately equal. The pressure uphole of the seal assembly **248** is the pressure inside the switch compartment **220** that is approximately atmospheric pressure. The pressure downhole of the seal assembly **248** is the pressure inside the setting tool **42** that is approximately atmospheric pressure. Therefore, the movable isolator **208** is pressure balanced.

(63) The ignition of the combustible element **49** inside the setting tool **42** by the igniter **246** will

produce high pressure gas. Turning now to FIG. 12 pressure within the setting tool 42 is greater than pressure within the switch compartment 220 which unbalances the movable isolator 208 and bias the movable isolator towards the switch compartment 220. For clarity, the inner bore 312 of the igniter retainer 204 is fluidly connected to the setting tool 42 and therefore the pressure within the setting tool 42 is also the pressure within the inner bore 312. The fluid pressure within the inner bore 312 urges the movable isolator 208 and seal assembly 248 towards the switch compartment 220. The movement of the movable isolator 208 within the igniter compartment 222 towards the switch compartment 220 compresses the retaining spring 244 and extends the insulated pin connector 242 into the switch compartment 220. The movement of the insulated pin connector 242 into the switch compartment 220 pushes the connector post 290 of the lower pin assembly 278 towards the spring retainer 294, compresses the connector spring 292, and moves the pin connector 288 into contact with the washer 316 of the grounding point assembly 314. The contact of the pin connector 288 of the lower pin assembly 278 to the washer 316 of the grounding point assembly 314 grounds the igniter switch 274. The grounding of the igniter switch 274 breaks communication with the surface personnel.

(64) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods may be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted or not implemented.

(65) Also, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component, whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

Claims

1. A system for plugging a wellbore extending from the surface and into a subterranean earthen formation, the system comprising: a plug comprising an annular sealing element and having an initial configuration configured to permit fluid flow around the plug in the wellbore and a set configuration configured to plug the wellbore whereby fluid flow around the plug in the wellbore is restricted; a setting tool configured to couple with the plug and comprising an energetic element received in a combustion compartment of the setting tool, the setting tool configured to shift the plug from the initial configuration to the set configuration in response to ignition of the energetic element; and a setting tool initiator comprising an addressable igniter switch and an igniter, wherein the igniter is configured to ignite the energetic element of the setting tool from a location that is external the combustion compartment in response to the igniter switch receiving a firing signal addressed to the igniter switch; wherein the igniter switch has an operational state in which the igniter switch is configured to receive electrical signals from the surface and an inoperable state in which the igniter switch is not configured receive electrical signals from the surface, and wherein the setting tool initiator is configured to shift the igniter switch from the operational state to the inoperable state in response to the ignition of the igniter.
2. The plugging system according to claim 1, wherein the igniter switch comprises a digitally addressable switch.
3. The plugging system according to claim 1, wherein the igniter switch comprises a processor and

a memory device including instructions stored therein defining the operation of the igniter switch.

4. The plugging system according to claim 1, wherein the setting tool initiator is configured to physically disable the igniter switch in response to the ignition of the igniter.

5. The plugging system according to claim 1, wherein the setting tool initiator is configured to expose the igniter switch to combustion products generated by the igniter in response to the ignition of the igniter.

6. The plugging system according to claim 1, wherein the setting tool initiator is configured to permanently shift the igniter switch from the operational state to the inoperable state.

7. The plugging system according to claim 1, wherein the setting tool initiator comprises an initiator housing having a switch compartment in which the igniter switch is positioned, and an igniter compartment in which the igniter is located, and wherein a fluid flowpath extends from the igniter compartment to the switch compartment.

8. The plugging system according to claim 1, wherein the setting tool initiator further comprises: a switch chassis having an interior in which the igniter switch is positioned; and an igniter adapter coupled to the switch chassis and having an interior in which the igniter is positioned.

9. A setting tool initiator for actuating a setting tool of a plugging system, the setting tool initiator comprising: an addressable igniter switch having an operational state in which the igniter switch is configured to receive and transmit electrical signals and an inoperable state in which the igniter switch is not configured to transmit or receive electrical signals; and an igniter ignitable by the igniter switch in response to the igniter switch receiving a firing signal addressed to the igniter switch, and configured to ignite an energetic element of the setting tool from a location that is external a combustion compartment of the setting tool in which the energetic element is received to actuate the setting tool in response to ignition of the igniter when the setting tool initiator is connected to the setting tool; wherein the setting tool initiator is configured to shift the igniter switch from the operational state to the inoperable state in response to the ignition of the igniter.

10. The plugging system according to claim 9, wherein the igniter switch comprises a digitally addressable switch.

11. The plugging system according to claim 9, wherein the igniter switch comprises a processor and a memory device including instructions stored therein defining the operation of the igniter switch.

12. The setting tool initiator according to claim 9, further comprising: an initiator housing in which the igniter switch and the igniter are received; a switch chassis positioned in the initiator housing and having an interior in which the igniter switch is positioned; and an igniter adapter positioned in the initiator housing and coupled to the switch chassis, wherein the igniter adapter has an interior in which the igniter is positioned.

13. The setting tool initiator according to claim 12, wherein the switch chassis positions the igniter switch at a predefined distance from the igniter.

14. The setting tool initiator according to claim 13, wherein the predefined distance is 1.75 inches or less.

15. The setting tool initiator according to claim 12, wherein the setting tool initiator further comprises an igniter spring positioned in the interior of the igniter adapter, wherein the igniter spring electrically connects the igniter switch with the igniter.

16. The setting tool initiator according to claim 9, further comprising: an initiator housing in which the igniter switch and the igniter are received, and wherein the initiator housing defines an igniter compartment in which the igniter is positioned and a switch compartment in which the igniter is positioned; wherein a fluid flowpath is formed extending through the initiator housing from the igniter compartment to the switch compartment.

17. The setting tool initiator according to claim 16, wherein the initiator housing has a maximum length of 6.5 inches or less.

18. The setting tool initiator according to claim 16, wherein the switch compartment has a maximum inner diameter of 1.50 inches or less.
